Does Social Media Effect Depression?

Aashiya Vahora, Minh Nguyen, Nathan Fernandez
QMST 4320 Data Analytics, McCoy College of Business, Texas State University

Emmett & Miriam McCoy College of Business Administration

Introduction

Social media's role is significant in contemporary society, acting as a primary source for global news, maintaining personal relationships, and fostering new social ties.

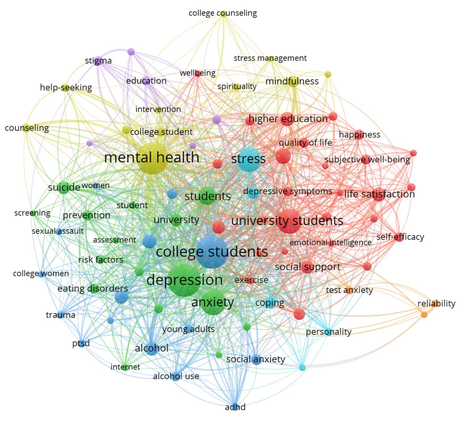
The research focuses on analyzing the effects of social media on depression, by examining survey data to understand social media's complex role in everyday life and mental well-being.

Another aspect of our study examines the interaction between university life and depression, aiming to determine how academic pressures impacts depression.

Motivation

Gain a clearer picture of the impact social media has on our mental health. As college students, we're immersed in a digital environment where social media is a constant presence.

We're curious about the real effects this has on our emotional well-being. Evaluating experiences of a wide range of people, we want to uncover patterns and insights that could help us all use social media in a way that's healthier.



Data & Methodology

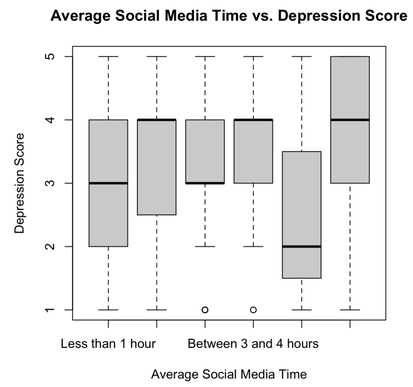
Ordinal Logistic Regression

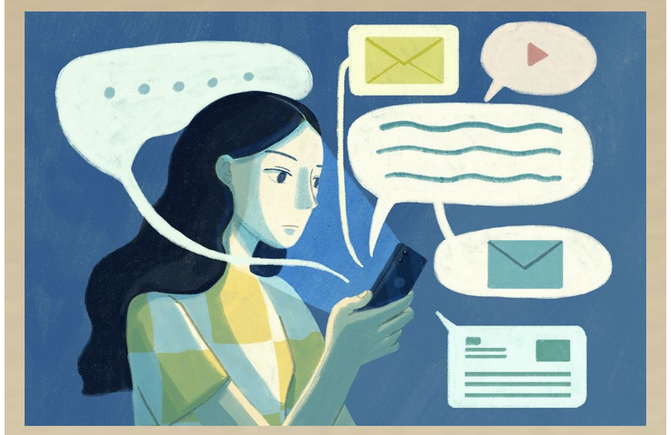
Data Set 1 Legend:

age: Respondent's age.
gender: Respondent's gender.
relationship_status: Respondent's relationship status.
occupation_status: Respondent's occupation status.
organization: Type of organizations the respondent is affiliated with.
socialmedia_use: Whether the respondent uses social media.
sm_platform: Social media platforms the respondent commonly uses.
sm_avgtime: The average time spent on social media every day.
sm_q1: How often the respondent uses social media without a specific purpose.
sm_q2: How often the respondent gets distracted by social media when busy.
sm_q3: Feelings of restlessness when not using social media for a while.
distracted_q: How easily the respondent is distracted (scale of 1 to 5).
bothered_q: How much the respondent is bothered by worries (scale of 1 to 5).
concentrate_q: Difficulty in concentrating on things.
selfcompare_q: How often the respondent compares themselves to other successful people through social media (scale of 1 to 5).
compare_q: Feelings about these comparisons generally speaking.
validation_q: How often the respondent seeks validation from social media.
depressed_q: Frequency of feeling depressed or down.
activites_q: Frequency of interest fluctuation in daily activities (scale of 1 to 5).
sleep_q: Frequency of facing issues regarding sleep (scale of 1 to 5).

Data Set 2 Legend:

Stress_Level: The level of stress experienced by the individual.
Depression_Score: The score representing the level of depression experienced by the individual.
Anxiety_Score: The score representing the level of anxiety experienced by the individual.
Sleep_Quality: The quality of sleep experienced by the individual .
Physical_Activity: The level of physical activity .
Diet_Quality: The quality of the individual's diet.
Social_Support: The level of social support received by the individual.
Substance_Use: The frequency of substance use such as alcohol, cigarettes or other drugs.
Family_History: Whether the individual has a family history of mental health issues.
Chronic_Illness:
Financial_Stress: The level of financial stress experienced by the individual (between 0 and 5).
Semester_Credit_Load: The number of credits the individual is taking in the semester (15-30).

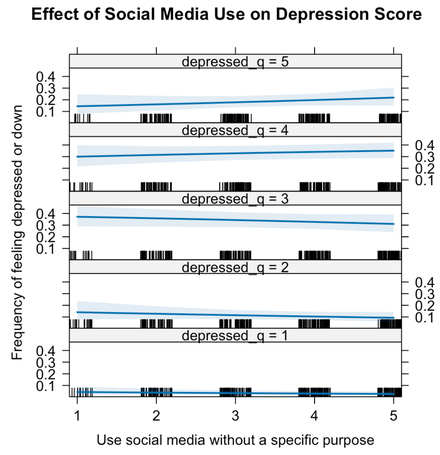


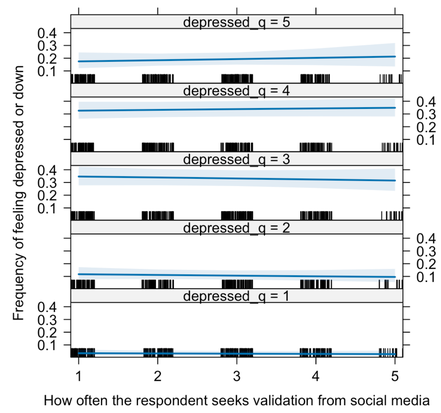


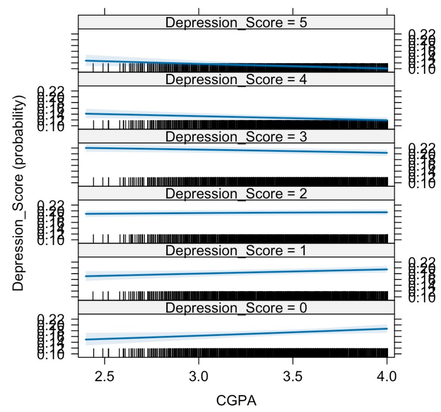
Results

Ordinal Logistic Regression Finding:

- More time spent on social media concludes a higher depression score among the survey participants.
- Social media without a purpose affects depression.
- More participants who compare themselves & seek validation from social media have a higher depression score.
- Our secondary data shows that with a higher GPA there is a lower probability of depression.







Conclusion

Our research shows that social media has become a kind of escape for many people, a place they go to avoid real-life problems.

However, our findings also show that spending more time on social media can actually make you feel more depressed.

We looked at other things that could affect how you feel, like stress from school or where you come from, social media still had a big impact on mental health.

References

https://www.kaggle.com/datasets/souvikahmed071/social-media-and-mental-health

https://www.kaggle.com/datasets/sonia22222/students-mental-health-assessments

Acknowledgements

This presentation is done as part of a project for QMST 4320 Data Analytics class, which is supervised by Dr. Tahir Ekin.

TEXAS STATE UNIVERSITY
The rising STAR of Texas